

Anti-jamming Path Planning for Unmanned Aerial Vehicles with Imperfect Jammer Information

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Abstract—In this paper, we consider an anti-jamming problem for one unmanned aerial vehicle (UAV) reaching a given destination with maintaining the communication quality, where the ground-to-air communication link is spatially affected by an adversarial jammer. Note that the jammer's position and transmit power are initially unknown. To solve this problem, we propose a jammer detection-localization-evasion framework. Firstly, we employ a jammer detection method combining the Received Signal Strength (RSS) with Packet Error Rate (PER). After the jammer is detected, our designed RSS based jammer localization algorithm is presented, which can estimate the jammer position and jamming signal power simultaneously. However, localization noises often exist in practical scenarios, and therefore the estimated jammer information is usually with uncertainties. To overcome these uncertainties, we propose a robust path planning method of successful jammer evasion. More specifically, the proposed method can guarantee the worst-case performance so that the communication quality can be always better than a certain level under all possible uncertainties. Finally, a UAV communication jammed model is built on Network Simulator (NS3) and corroborates the effectiveness of our proposed method.

Index Terms—Unmanned aerial vehicles, imperfect jammer information, jammer localization, robust path planning

I. INTRODUCTION

A. Motivation and Related Work

Due to good maneuverability, convenient operation and autonomy, unmanned aerial vehicles (UAVs) have gained rapid development in a wide range of applications, ranging from package delivery, environmental monitoring to search-and-rescue tasks. UAVs adopt wireless communication technologies to build up the data link to Ground Control Station (GCS), including 3G/4G/5G mobile network, WiFi, ZigBee and so on. The shared nature of the wireless network makes UAVs vulnerable to intentional or unintentional attacks. As a kind of malicious attack, the jammer interrupts normal communication by emitting radio signal to cause collision at the receiver. Once being attacked, the quality of communication will decline, and even fails to meet the current demand and cause the mission interrupted. Hence, in order to ensure the completion of mission, there are two main problems that need to be

clarified: 1) whether jammer exists and where it is; and 2) how to defend against jammer to reduce its impact on mission.

A series of strategies coping with jamming attack have been proposed in wireless networks, which are generally divided into two categories. One is to retreat away from the jammer, the other is to adapt to the jamming signal. As for retreat, in [1], Xu et al. proposed two evasion approaches: channel surfing and spatial retreats. Channel surfing involved legitimate wireless nodes changing communication channel, and spatial retreats belongs to spatial evasion where the jammed node selects randomly the direction of movement until escaping from the jammed area. As for competing with jammer, Lv et al. [2] presented that legitimate nodes adjust transmit power level to make receivers get higher signal-to-interference-plus-noise ratio (SINR). In [3], a stronger error correcting code with a lower information rate was employed to reduce the overall throughput but increase the likelihood of packets successful being decoded.

Since UAVs have better mobility and flexibility than the traditional nodes in wireless networks, it is reasonable and suitable to consider spatial evasion to cope with jamming attacks. Furthermore, spatial evasion based methods do not have high requirements of communications devices. In [4], a differential game theoretic approach was designed to compute optimal flying strategy to evade from the attack of an aerial jammer. But whether the communication channel was considered to be jammed, was simply judged by whether the ratio of transmitter-receiver distance and jammer-receiver distance is higher than a predefined value. Judd et al. [5] developed a trajectory planning scheme generated from a Voronoi diagram, whose edge was assigned to a threat cost, which was inversely proportional to the fourth power of the distance between the receiver and the threat. Duan et al. [6] provided a more accurate description about jamming effectiveness based on analytic hierarchy process (AHP), which combined various factors including the radar relative position and signal power.

The aforementioned spatial retreat approaches highly rely on specific information (like the location and power information) of jammers, which can hardly be known in advance. Moreover, these approaches did not consider the communication model for spatial retreats, which may result in the loss of analytical accuracy. Indeed, jammer's effect on the receiver is the result of a complicated communication process. The effect

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of jamming is not only related to the position of the transmitter and jammer, the transmit power, but also path loss function. Hence, detecting jammer and solving jammer position are the prerequisite of spatial retreats.

There have been a lot of works in jammer detection and localization. Generally speaking, jamming detection techniques relied on the received signal strength (RSS), packet delivery ratio (PDR) or the utility of a channel as their main decision criteria [7]–[10]. In terms of jammer localization, [11] proposed a localization algorithm based on hearing range (the area where nodes can receive packets successfully). Pelechrinis et al. [12] adopts gradient descent search to locate the jammer by measuring PDR changes. However, the above jammer detection and localization schemes cannot be employed to spatial retreats of UAVs, since: 1) the position of a UAV changes over time, which makes the jammer detection and localization more difficult; and 2) the jammer often has a different altitude from the UAV, and thus the jammer localization must be considered in 3-D space (rather than 2-D space in conventional problem settings).

To sum up, it is very necessary and meaningful to study the UAV anti-jamming problem: by combining the jammer detection, localization, and evasion (or spatial retreat) altogether; and by considering the unique properties of UAVs.

B. Our Contributions

In this work, we study an anti-jamming problem for one UAV to reach a given destination while maintaining the communication quality as well as minimizing the communication loss, where the ground-to-air communication link is spatially affected by an adversarial jammer. Note that the jammer's position and transmit power are initially unknown. To solve this problem, we propose a jammer detection-localization-evasion framework.

The main contributions are as follows:

- We design a RSS based jammer localization algorithm in 3-D space, which can simultaneously determine the range of position and transmit power of the jammer.
- With considering the localization uncertainties, we propose a robust path planning method to guarantee the worst-case performance so that the communication quality can be always better than a certain level under all possible uncertainties.

C. Paper Organization

The rest of the paper is organized as follows. In Section II, we present a wireless transmitter-receiver model in the presence of jammer and formula the problem. In Section III, we detail anti-jamming method for UAVs and processes onboard. The simulation results are given in Section IV to test the effectiveness of our approach. Finally, conclusion is drawn in Section V.

II. SYSTEM MODEL AND PROBLEM DESCRIPTION

A. System Model

Consider a ground-to-air UAV communication link with an adversarial jammer which attempts to interrupt the transmis-

sion (see Fig. 1). Tx and Rx denote the transmitter and the receiver, respectively, and J indicates the jammer. The triple (x_T, y_T, z_T) represents the position of Tx and (x_R, y_R, z_R) depicts the position of Rx. Jammer's position is (x_J, y_J, z_J) . The transmitter Tx sends data packets to the receiver Rx periodically. P_T determines the transmit power of the transmitter Tx. Meanwhile, the jammer J sends high power signal at the same frequency range to cause data transmission failure, and its power is P_J . Furthermore, position and transmit power of jammer are unknown for the UAV in advance.

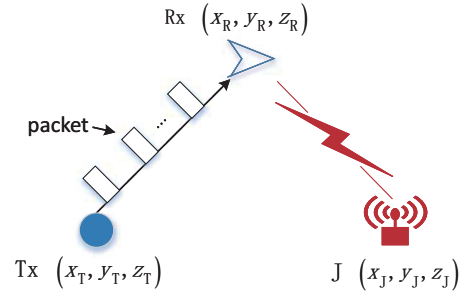


Fig. 1. A wireless transmitter-receiver pair in the presence of a jammer. The solid arrow and lightning line indicate the desired data delivery and the jamming link, respectively.

Radio signal will be dispersed as the propagation distance increases. From the position coordinates of the transmitter and receiver, we get the distance between them

$$r_{TR} = \|(x_T, y_T, z_T) - (x_R, y_R, z_R)\|. \quad (1)$$

At this time, we only consider omnidirectional antenna which radiates electromagnetic waves in spherical form. According to path loss effect, the relationship between the transmit power with the received power is as follows:

$$P_{TR} = P_T \cdot r_{TR}^{-\alpha}, \quad (2)$$

where P_T is the transmit power strength, P_{TR} is the power strength reaching the receiver, α is path loss exponent, generally $\alpha \in [2, 4]$. The received jamming signal P_{JR} can be derived in a similar way. The Received Signal Strength (RSS) of the receiver Rx consists of three aspects, including P_{TR} , P_{JR} and noise P_N . Hence, RSS is denoted as

$$RSS = P_{TR} + P_{JR} + P_N. \quad (3)$$

When a packet arrives at the receiver, the receiver checks the preamble to discern whether it is valid or jamming packet. Jamming packet is regarded as interference, which causes the decrease of signal-to-interference-plus-noise ratio (SINR). The SINR is determined by

$$SINR = \frac{P_{TR}}{P_{JR} + P_N}. \quad (4)$$

In digital modulation system, Bit Error Rate (BER) depends on signal-to-noise of demodulator. Here, the jamming signal

is also be regarded as noise for the receiver. The higher SINR, the lower BER. BER is defined as

$$BER = f(SINR), \quad (5)$$

where, f is a decreasing function, which is determined by signal modulation and demodulation method. Packet Success Rate (PSR) represents the probability of receiving a packet correctly. One packet contains lots of bits, and PSR is obtained by multiplication of all bit accuracies, represented as

$$PSR = (1 - BER)^{nbits}, \quad (6)$$

where $nbits$ represents the number of bits in this packet. Packet Error Rate (PER) is the inverse of PSR,

$$PER = 1 - PSR. \quad (7)$$

Generally, we choose RSS and PER as the basic statics to detect the presence of jammer and estimate jammer's position. PER is adopted to analyze whether communication quality can meet mission requirement.

B. Problem Description

Consider a situation in which a UAV is executing one mission in the specific region. As described in Fig. 2, a UAV starts from the source point S and flies to the destination point T . Meanwhile, it needs to keep wireless communication with GCS, which sends data packets containing control instruction or other mission related information to the UAV periodically. If the UAV is attacked by a jammer, PER rise sharply in some areas resulting in that these packets fail to be decoded. The red area in Fig. 2 is severely jammed, where PER can not satisfy mission requirement. Generally, when the UAV finds it is jammed, it usually flies back or hover. Because the communication quality of the front area can not be predicted.

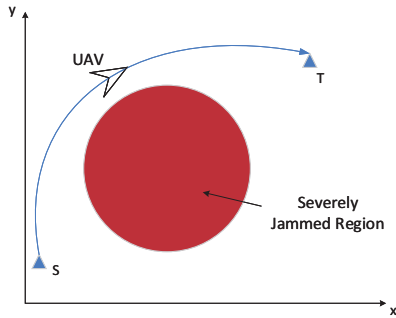


Fig. 2. A UAV evades from the severely jammed region to reach the destination.

Therefore, this paper focuses on how to ensure that a UAV can maintain communication quality to complete current mission in the presence of a jammer, which consists of the following subproblems:

- 1) Judge whether the UAV is attacked by a jammer;
- 2) Estimate the range of jammer position and its transmit power;

- 3) Analyze communication quality and plan a robust path which meets mission requirement for communication although in the worst case.

III. PROBLEM SOLUTION

In this section, we will give detailed discussions on the main process of our solution. As shown in Fig. 3, firstly, the UAV monitors the changes of RSS and PER onboard to detect timely the presence of jammer, here, we adopt the consistency check combining RSS with PER. Then, if there is a jammer, the UAV starts to collect RSS in different positions to locate jammer in three-dimensional space. Due to noise influence, it only estimate the range of jammer's position and transmit power, which will generate a series of corresponding PER distribution. Next, the communication quality should be analyzed by PER to determine whether the current path will meet mission requirement. If not, the current path need to be adjusted. Lastly, we develop an robust path planning algorithm based on max PER distribution with path length constraint. The whole process will run onboard for an autonomous UAV to cope with the jammer.

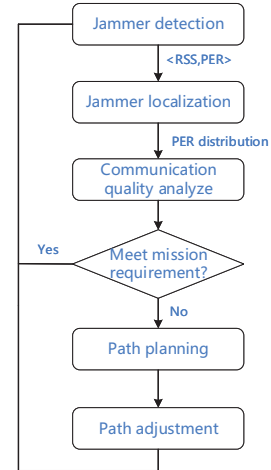


Fig. 3. UAV workflow

A. Jammer Detection

We consider a UAV records RSS and PER while performing its mission. One powerful measurement employed to detect jammer is PER, the distribution of which will change greatly due to the jammer attack. When receiver reaches the area close to jammer, the received jamming signal power will increase sharply, which causes the decrease of SINR. Then PER will rise with it. Hence, the closer to jammer J , the higher the PER, as shown in Fig. 4(a). Whereas, there are still other factors that make PER increases, such as poor link quality. In order to distinguish with this situation, we add RSS as additional measurement. In normal condition, high PER usually corresponds to low RSS. Conversely, in Fig. 4(b), a jammer transmits radio signal, leading to high RSS. Therefore, the UAV can prove the presence of jammer, when finding the increase in PER and RSS at the same time.

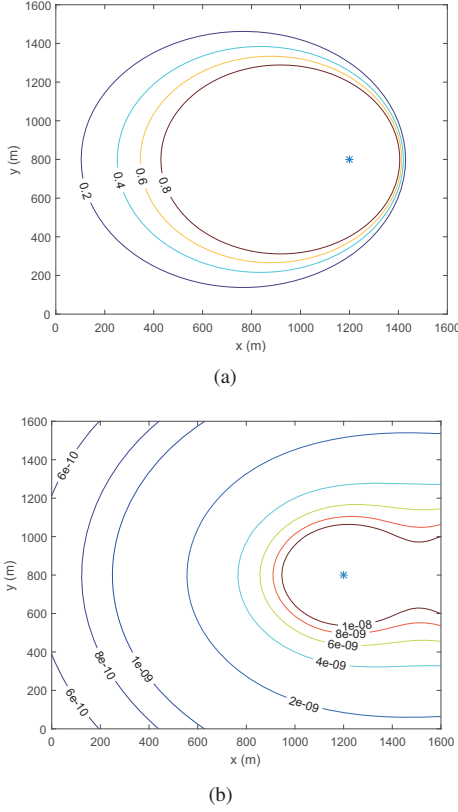


Fig. 4. PER (a) and RSS (b) distribution in presence of a jammer. It works on 915MHZ frequency, the transmit signal power of the transmitter and jammer is 0.9watt and 1watt, respectively.

B. Jammer localization

Radio wave will not only be dispersed as the propagation distance increases, but also is influenced by many other factors including terrain and weather. According to (2), the UAV can calculate the P_{TR} . Also, RSS can be gotten in real time. By subtracting P_{TR} from RSS, we can calculate the received jamming signal P_{JR} with the noise P_N , which is represented by

$$P'_{JR} = P_{JR} + P_N. \quad (8)$$

The level of P_N can not be determined, for which we set a range,

$$\Delta P_N = [-N_{max}, +N_{max}], \quad (9)$$

where, N_{max} denotes noise boundary, by which we can get a scope of P_{JR} ,

$$P_{JR} \in [P'_{JR} - N_{max}, P'_{JR} + N_{max}]. \quad (10)$$

We propose a jammer localization and transmit power solution method based on P_{JR} in three-dimensional space. Only having a rough range for P_{JR} , we get a series of corresponding jammer positions and power. As shown in Fig. 5, jammer position is (x_J, y_J, z_J) , which does not change over time. Its transmit power is P_J . The position of the UAV at time t is

denoted as $(x_R(t), y_R(t), z_R(t))$, and the received jamming signal power is $P_{JR}(t)$. Fig. 5 shows the UAV positions and the received power at t_1, t_2, t_3 . Based on the relationship between

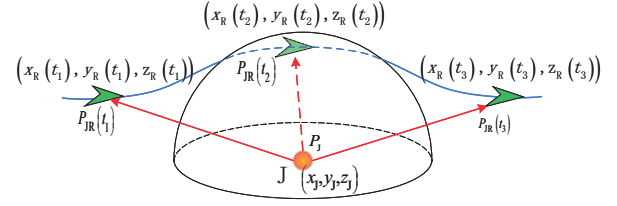


Fig. 5. The UAV samples the received jamming signal power on its three-dimensional motion trajectory.

distance and signal attenuation (2), we convert propagation loss to the distance between UAV and jammer. It can be represented as

$$(x_R(t) - x_J)^2 + (y_R(t) - y_J)^2 + (z_R(t) - z_J)^2 = \left(\frac{P_J}{P_{JR}(t)} \right)^{\frac{2}{\alpha}}. \quad (11)$$

The UAV samples the position and received jamming power of n moments, and puts them into (11). let these n moments be $(t_1, t_2, \dots, t_n)$, and by subtracting the equations at the two adjacent sampling moments, a new set of equations is formed as (12). When the non-coplanar sample points reach five, solution will uniquely exist.

$$[A; B]X = C, \quad (12)$$

where,

$$X = \begin{bmatrix} x_J & y_J & z_J & P_J^{\frac{2}{\alpha}} \end{bmatrix}^T,$$

$$A = \begin{bmatrix} 2x_R(t_1) - 2x_R(t_2) & 2y_R(t_1) - 2y_R(t_2) & 2z_R(t_1) - 2z_R(t_2) \\ \vdots & \vdots & \vdots \\ 2x_R(t_{n-1}) - 2x_R(t_n) & 2y_R(t_{n-1}) - 2y_R(t_n) & 2z_R(t_{n-1}) - 2z_R(t_n) \end{bmatrix},$$

$$B = \begin{bmatrix} P_{JR}^{-1}(t_1) - P_{JR}^{-1}(t_2) \\ \vdots \\ P_{JR}^{-1}(t_{n-1}) - P_{JR}^{-1}(t_n) \end{bmatrix},$$

$$C = \begin{bmatrix} x_R^2(t_1) - x_R^2(t_2) + y_R^2(t_1) - y_R^2(t_2) + z_R^2(t_1) - z_R^2(t_2) \\ \vdots \\ x_R^2(t_{n-1}) - x_R^2(t_n) + y_R^2(t_{n-1}) - y_R^2(t_n) + z_R^2(t_{n-1}) - z_R^2(t_n) \end{bmatrix}. \quad (13)$$

C. Communication quality analysis

The wireless communication quality is a vital factor in determining whether UAV can complete mission. Various types of missions have different requirements for communication quality. For instance, remote control flying requires high delivery rate for control instructions. However, when attacked by a jammer, the data link is blocked to cause mission quality. During the execution of missions, the UAV monitors communication condition to judge whether it satisfy mission requirements.

D. Robust Path Planning

If it is found that the original path cannot meet mission requirements, the next step to consider is how to avoid these severely interfered areas and fly to destination as quickly as possible. We adopt PER as a criteria for communication quality, where high PER means poor communication quality. That is, PER of the whole path need to be low than the PER limit of missions. An robust path planning model based on PER distribution with a constraint on the path length is developed.

1) *Model Description:* Since the location and transmit power of the jammer is only estimated for a range of values, we can only get a series of possible PER distributions. The objective of the path planning model is to optimize the worst-case performance.

The UAV executes missions on the same height plane, It changes flying height slightly in locating jammer process and then return to the original height. Therefore, the path planning is established on two-dimensional plane. It is formulated in the following way. The cost function $R(PA)$ is PER accumulated along the planning path PA . We use I_J to represent jammer's information including jammer's position (x_J, y_J, z_J) and jammer's transmit power P_J . $per(x, y, I_J)$ is PER at the position (x, y) when jammer is in the state of I_J . ds denotes the per length. S_p and T are the start point and destination point of planning path PA , respectively.

There is a constraint on PER that can not be ignored. When $R(PA)$ is at a minimum, there is possibly one or several waypoints whose PER is high enough to make UAV lose communication ability halfway. So we need to set a threshold $\overline{per}$ depended on UAV mission, PER of any waypoint must be lower than the threshold. Moreover, we set a planning path length limit $\bar{l}$ to prevent the phenomenon that the UAV go very far to areas of extremely low PER, which is not economic because mission generally can tolerate a certain number of error packets. The Robust-Path-Planning game is given by the following expression

$$\begin{aligned} \min_{PA} \max_{I_J} \quad & R(PA) = \int_{S_p}^T per(x, y, I_J) ds \\ \text{subject to} \quad & per(x, y, I_J) \leq \overline{per}, \\ & l(PA) = \int_{S_p}^T ds \leq \bar{l}. \end{aligned} \quad (14)$$

Remark 1. We transform the model into a game between jammer's position, transmit power and the UAV, where the former is to choose a PER distribution to maximize PER accumulation, but the latter is to choose a path to minimize it.

2) *Model Solution:* It should be noted that the optimal solution to Robust-Path-Planning game (14) is difficult to derive. We use the relaxation method to calculate the upper bound of the optimal solution, which can guarantee performance. Due to noise's influence on jammer localization, we get lots of possible PER distributions. By comparing all these PER

distributions, we get max PER of every node to form a max PER distribution, denoted by the matrix $maxper$. On the basis of $maxper$, the objective is to get a path with the minimal accumulated PER while meeting constraints. The optimization problem is described as

$$\begin{aligned} \min_{PA} \quad & \int_{S_p}^T maxper(x, y) ds \\ \text{subject to} \quad & maxper(x, y) \leq \overline{per}, \\ & l(PA) = \int_{S_p}^T ds \leq \bar{l}. \end{aligned} \quad (15)$$

For (14), we give a certain path PA ,

$$\max_{I_J} R(PA) = \int_{S_p}^T per(x, y, I_J) ds. \quad (16)$$

Assuming $R_{max}(PA)$ is the max value of (16) and I_J^* is the corresponding jammer information,

$$R_{max}(PA) = \int_{S_p}^T per(x, y, I_J^*) ds. \quad (17)$$

Because $maxper$ is the max PER matrix,

$$maxper(x, y) \geq per(x, y, I_J), \quad (18)$$

$$\int_{S_p}^T maxper(x, y) ds \geq R_{max}(PA). \quad (19)$$

It proves that the objective value obtained by optimization problem (15) is an upper bound of the game (14), which can guarantee the performance of the system. Discrete optimization method is adopted to solve problem (15). The flight area is evenly divided into small grids, where each vertice is one possible UAV waypoint. To solve discrete optimization with constraints, modified label-setting algorithm [13] is employed. The detailed solution steps are illustrated in Algorithm 1.

Algorithm 1 Solve robust path planning model

Input: A series of jammer's positions and transmit powers, $I_J(x_J, y_J, z_J, P_J)$; The planning source and destination point, S_p, T ; The planning path length limit, $\bar{l}$; PER limit of mission, $\overline{per}$;

Output: The optimal planning path, PA^* ;

- 1: Discretize the flight area to form an undirected fully connected graph.
 - 2: Calculate lots of PER distributions based on I_J to get max PER matrix $maxper$;
 - 3: Delete nodes whose PER is higher than $\overline{per}$;
 - 4: Use modified label-setting algorithm [13] to solve PA^* . $S_p, T, \bar{l}$ and PER of every graph node are inputs of this algorithm.
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In Algorithm 1, the modified label-setting algorithm uses a set of labels for each node. These labels correspond to different paths from source point to this node, each is composed of accumulated PER and path length. Labels with lower accumulated PER have longer path. Hence, we need to get

destination point's label with minimal accumulated PER while satisfying path length limit. $\bar{l}$.

IV. SIMULATION RESULTS

In this section, we present simulation results from jammer detection, jammer localization to path planning. Using Network Simulator (NS3), the communication states of UAV and GCS in the presence of a jammer are shown. NS3 can record RSS and PER, which are the basis of subsequent experiment.

Assuming one UAV performs mission within an area of $1.8\text{km} \times 1.8\text{km}$, its flying height almost remains 100m. The UAV starts from the source point $(0, 0, 100)$ and flies directly to the destination point $(1600, 1600, 100)$ at a speed of 10m/s. Meanwhile, it communicates with GCS at $(1800, 800, 0)$ by wireless network. In NS3, we assume this communication process at 915MHZ frequency. GCS sends data packets of 200bytes to the UAV at 0.9watt power, at this time, a jammer also sends signal to interrupt normal communication at 915MHZ frequency. The jammer is set at $(1200, 800, 0)$ far away from the source point and transmit power is 1watt. All information about jammer is initially unknown to the UAV.

A. Jammer detection

From the the source point $(0, 0, 100)$, the UAV monitors RSS and PDR changes in real time. After about 40 seconds, it arrives at point $(280, 280, 100)$. RSS was found to increase to 8.406×10^{-7} watts and PER rapidly from 0.02 to 0.15 in Fig. 6. According to jammer detection method, the phenomenon that RSS and PER increase in the same time can prove the UAV is attacked by a jammer.

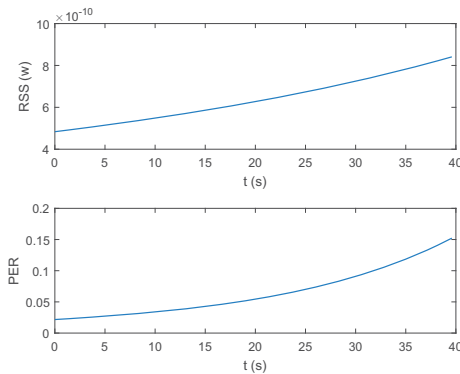


Fig. 6. PER (a) and RSS (b) changes until detecting jammer

B. Jammer Localization

After the jammer has been detected, the UAV immediately activates jammer localization algorithm. It starts to move around. In order to uniquely identify the jammer location, samples need to be non-coplanar, which means that the UAVs height has small changes. It collects RSS values of 5 sample points, from which are subtracted the corresponding P_{TR} to get P'_{JR} according to (8). Here, we set noise as $P_N \in [-10^{-11}, 10^{-11}]$, and the UAV substitutes P'_{JR} into the (10) to get the scope of P_{JR} . According to (12), the scopes of

jammer's positions and transmit powers are solved. As shown in Fig. 7, estimated jammer's positions form roughly a line from $(1153, 773.6, 5.359)$ to $(1251, 828.4, -5.773)$ and get farther from sample points as noise becomes greater. Also, transmit power increases with noise.

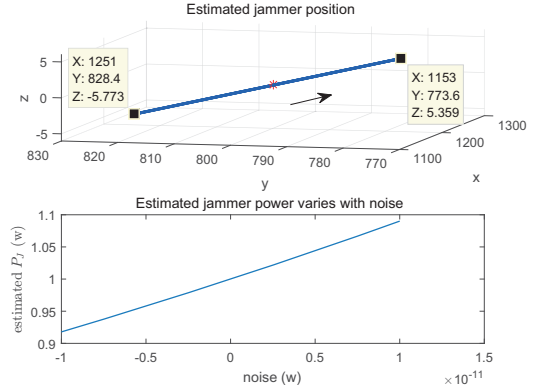


Fig. 7. The estimated positions and powers of jammer. The arrow points in the direction of increased noise.

C. Robust Path Planning

The UAV reaches the point S_p at $(300, 300)$ and continues to move in two-dimensional plane of 100 meter height. The region is evenly divided into lots of grids, the area of which is $100\text{m} \times 100\text{m}$. The position and signal transmit power of jammer has been estimated in Section IV-B, moreover, relevant information of GCS is all known in advance. Max PER distribution matrix $maxper$ will be gotten, then the UAV starts to replan a path which meet mission requirement.

As shown in Fig. 8, the solid equipotential line shows max PER distribution, whereas, the dotted equipotential line is the PER distribution without noise influence and jammer localization is accurate. We can see that max PER distribution covers PER distribution without noise influence. We set two PER limits $\overline{per} = 0.4$ and $\overline{per} = 0.6$, which correspond to two kinds of mission (rescue and delivery), respectively. Rescue task requires real-time return of environmental information, which has higher requirements for communication quality. We choose path length limit $\bar{l}$ as 2600, for the first scenario $\overline{per} = 0.4$. When PER limit is higher, the area whose communication quality does not meet mission requirement will be smaller. Hence, for the second scenario $\overline{per} = 0.6$, we choose path length limit $\bar{l}$ as 2400. The path1 and path2 in Fig. 8 bypass the area where $\overline{per}$ is higher than 0.4 and 0.6, respectively.

Then, we change the noise range with other settings unchanged. We choose per limit as $\overline{per} = 0.4$, and path length limits as $\bar{l} = 2600$. As Fig. 9 shows, we set twenty kinds of noise range, and noise boundary N_{max} ranges from 0 to 2×10^{-11} . For every noise range $[-N_{max}, N_{max}]$, we get a path, which is then used to calculate accumulated PER. It can be found that accumulated PER based on $maxper$ (guaranteed bound) is higher than accumulated PER of worst case in

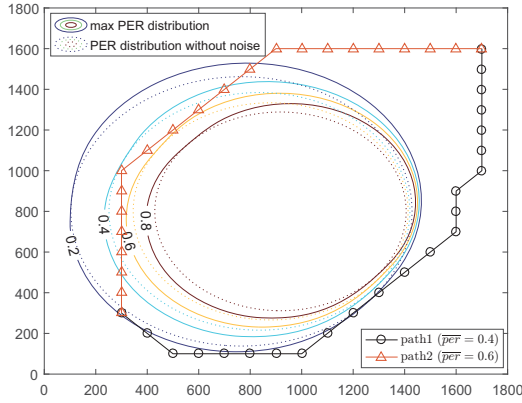


Fig. 8. Planning path and PER distributions.

this noise range. Hence, path planning on the basis of max PER distribution is feasible. Furthermore, the gap between guaranteed bound and worst-case PER get larger as noise range becomes higher. Whereas, the max gap only accounts for 10%.

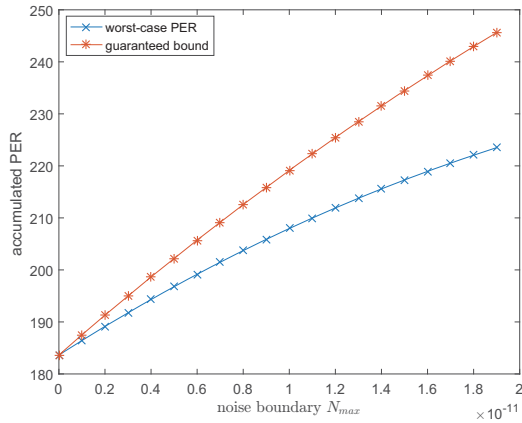


Fig. 9. Accumulated PER on max PER distribution $maxper$ and the worst-case PER distribution under different noise range

V. CONCLUSION

In this paper, an anti-jamming problem for an UAV with maintaining the communication quality has been studied. Jammer location and transmit power are initially unknown. We proposed a jammer detection-localization-evasion framework to solve this problem. We adopted a jammer detection method combining RSS with PER. We designed RSS based jammer localization algorithm to estimate the jammer position and jamming signal power simultaneously. With considering the localization uncertainties due to noise, we provided a robust path planning method of successful jammer evasion to minimize communication loss. This method is based on max PER distribution, which guarantees the worst-case performance.

More importantly, the presented work serves as the first step to develop more comprehensive results on anti-jamming problem for UAVs in the future:

- It is worth trying to locate several jammers, because the UAV is possibly attacked by more than one jammer in practical scenarios.
- It would be desirable to analyze how multiple UAVs in the formation cooperate to cope with jammers.

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